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Paleomagnetic evidence for a post-Cretaceous accretion of Wrangellia

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ABSTRACT

In order to test the various estimates of the accretion age of some of the southern Alaska terranes, a paleomagnetic investigation of the MacColl Ridge Formation (latest Cretaceous age) in the Wrangell Mountains (Wrangellia) was undertaken. The data pass the fold test and reversal test, thus affirming the primary nature of the magnetic remanence. The mean paleomagnetic direction corresponds to a paleolatitude of 32° ±9°, which is far south of the predicted paleolatitude, assuming a North American paleogeography. Therefore, these results apparently rule out a pre-Cenozoic accretion age of Wrangellia and its associated terranes composing the Southern Alaska superterrane. The data also imply about 130° counterclockwise rotation, which is the opposite sense of rotation usually observed in western North America.

INTRODUCTION

It has become increasingly apparent that southern Alaska is allochthonous with respect to North America (Jones et al., 1977; Coney et al., 1980; Jones et al., 1983). The general tectonostratigraphic framework has been identified by Jones and Silberling (1979) and Jones et al. (1981), and paleomagnetic data have demonstrated that at least some of these terranes occupied low latitudes in pre-Cenozoic time.

Packer and Stone (1972) first presented paleomagnetic evidence suggesting that the Peninsular terrane was south of its present location in Jurassic time. Hillhouse (1977) determined a Triassic paleolatitude of 15° for Wrangellia, far south of its present location with respect to North America. Since the initial recognition of allochthonous paleomagnetic signatures, several subsequent studies have confirmed the low Mesozoic paleolatitudes for southern Alaska (Stone and Packer, 1979; Stone, 1982; Hillhouse and Gromme, 1984).

Regional geologic studies suggest that many of these southern Alaska terranes amalgamated to form a superterrane prior to accretion (Berg et al., 1972). Csejtsy et al. (1982) employed the name "Talkeetna superterrane" to describe the amalgamation of southern Alaska terranes. However, I (Panuska, 1984a, 1984b) prefer the name "Southern Alaska superterrane" (comprising the Peninsular, Wrangellia, Alexander, and Gravina-Nutzotin terranes; see Fig. 1) on the basis that the Talkeetna superterrane does not include the Alexander terrane, which is an integral part of the amalgam (Berg et al., 1972). Although the exotic nature of the Southern Alaska superterrane is generally accepted, there is little agreement on the time of accretion of these terranes to North America.

Recent estimates of the accretion age of the Southern Alaska superterrane range from Early Cretaceous to middle or late Tertiary. Jones et al. (1982a) suggested an Early Cretaceous date of emplacement based on the probable time of deformation in the eastern Alaska Range and Wrangell Mountains. On the basis of structural and stratigraphic data from the central Alaska Range, Jones et al. (1982b) suggested a post-Cenomanian (Late Cretaceous), pre-Paleocene accretion. Early Tertiary paleomagnetic data from Stone (1984) yield a low paleolatitude and suggest a middle or late Tertiary accretion. However, Eocene paleomagnetic data reported by Hillhouse and Gromme (1983) seem to indicate that at least part of southern Alaska was in place with respect to North America by Eocene time. In order to provide more stringent constraints on the date of emplacement of Wrangellia and, by implication, the Southern Alaska superterrane, a paleomagnetic study was undertaken in the southern Wrangell Mountains, Alaska.

PALEOMAGNETIC DATA

The MacColl Ridge Formation (Maastrichtian) was sampled along two limbs of a syncline. The limbs occupy approximately the same stratigraphic position. In general, two cores were drilled from each of 15 beds in the north limb of the syncline (WRG.21) and 11 beds in the south limb (WRG.22). Specimens

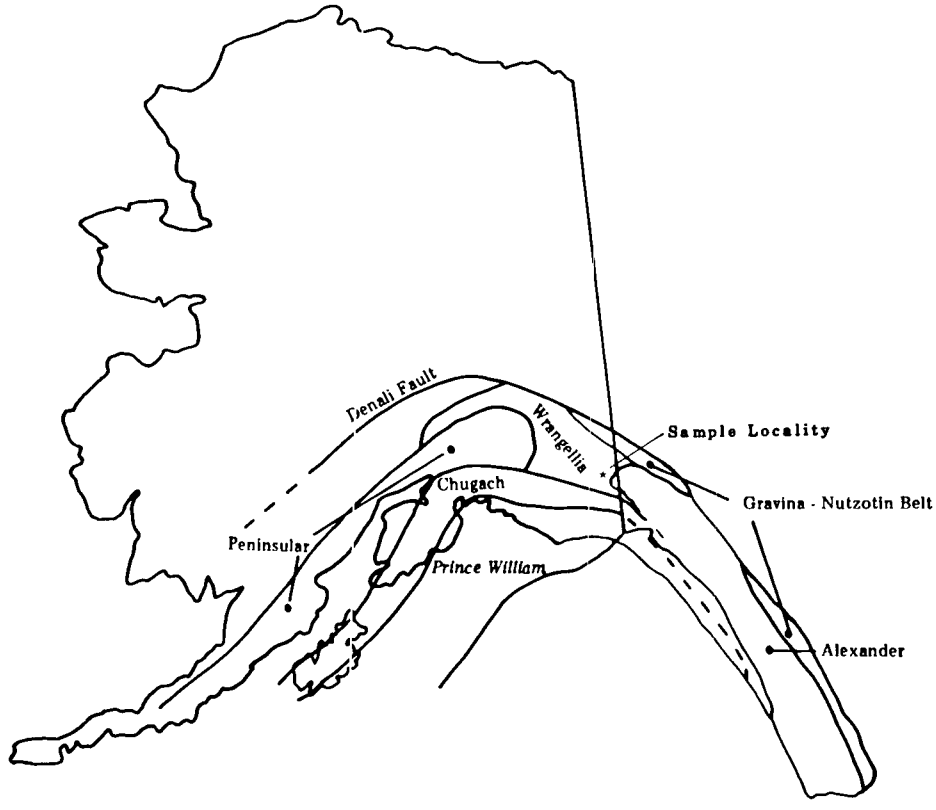


Figure 1. Generalized location map of major tectonostratigraphic terranes occurring south of the Denali fault (after Jones et al., 1981).

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cut from these cores were magnetically cleaned by thermal and/or alternating field (AF) techniques so as to remove the obscuring effects of secondary components of magnetization. The details of sampling and measurement are given in Panuska (1984b).

Demagnetization curves for most of the 30 thermally demagnetized specimens displayed intensity "spikes" at temperatures of 250 to 350 °C, suggesting that heating drove chemical reactions which altered the magnetic properties of the rocks. These specimens behaved erratically, and no stable magnetic directions could be isolated.

Well-defined directions of magnetization (characteristic direction) were, however, isolated from the AF and thermal/AF demagnetized specimens (thermal demagnetization to 180 °C followed by AF demagnetization). These directions were determined on the basis of removal of secondary magnetic components, demagnetization diagrams showing a decrease in intensity with little or no change of direction over successive cleaning steps, and lack of any significant rotational remanent magnetization (RRM; see Stevenson, 1980) overprinting induced during AF cleaning. About 25% of the AF and thermal/AF cleaned specimens satisfied these criteria.

Figure 2 shows the AF demagnetization behavior of three representative specimens. Specimens A and B show the removal of secondary components of magnetization by AF demagnetization at 100 to 150 Oe. Better characterization of the high coercivity component of magnetization was prevented by a strong RRM imparted to the specimen at 300 Oe (not shown for the sake of clarity); however, the coercivity spectra of the other specimens suggest that there would be little change in the magnetic vector if RRM-free measurements could be obtained at higher AF intensities. Specimen C shows a straight line demagnetization trajectory toward the origin, indicating that there is no secondary magnetic component.

Stability tests (fold test and reversal test) were applied to the characteristic directions of magnetization listed in Table 1 in order to determine whether the data represent primary remanence directions, which date back to the origin of the rock. This is especially critical in view of the large number of samples that did not clean up to directionally stable end points. These data pass the fold test (Fig. 3) at 95% confidence, using the test of McFadden and Jones (1981). This demonstrates that magnetization was acquired before folding. The age of folding is not well constrained in the Wrangell Mountains (post–latest Cretaceous, pre-Miocene) but is probably Eocene/Oligocene or older (Little, 1984).

The occurrence of magnetic reversals further strengthens the argument for a primary remanence. These reversals are stratigraphically controlled (Panuska, 1984b), which strongly suggests that they are penecontemporaneous with deposition. Although the mean normal and reversed magnetic directions are statistically antiparallel at 95% confidence, it is noted that the inclination of the reversals are shallower than the normal polarity data. This could be due to a secondary component of magnetization that was not completely removed by AF cleaning. The statistically antipodal normal and reversed direction indicate that if an unremoved secondary component is present in these data, it does not cause a significant deviation in the mean.

When dealing with sedimentary rocks, there is always some concern that the measured primary remanence direction accurately reflects the original geomagnetic field direction. This is because some sediments contain a magnetic inclination that is systematically lower than the magnetizing field, which is related to the depositional process. This systematic lowering of the recorded field is known as the inclination error.

A review of the literature on sediment magnetization processes by Verosub (1977) indicates that inclination errors are relatively uncommon. Payne and Verosub (1982) have studied the magnetization of natural sediments under laboratory controlled conditions. They found that the magnetic carriers in sands with less than about 40% matrix could reorient along a new field direction so long as the sediment water content was sufficiently high. This suggests that relatively well sorted sands could have their magnetic carriers realign with the geomagnetic field immediately after deposition and then become immobilized by compaction and dewatering, thus "erasing" a potential inclination error. Because the MacColl Ridge Formation specimens consist of well-sorted, fine- to medium-grained sandstones, the existence of a significant inclination error is improbable. Therefore, the combined mean direction of 226° declination, 51° inclination, K = 20.3, α_{95} = 9.1°, N = 14/20 (number of beds/number of specimens) is considered to be a statistically valid estimate of the ancient geomagnetic field.

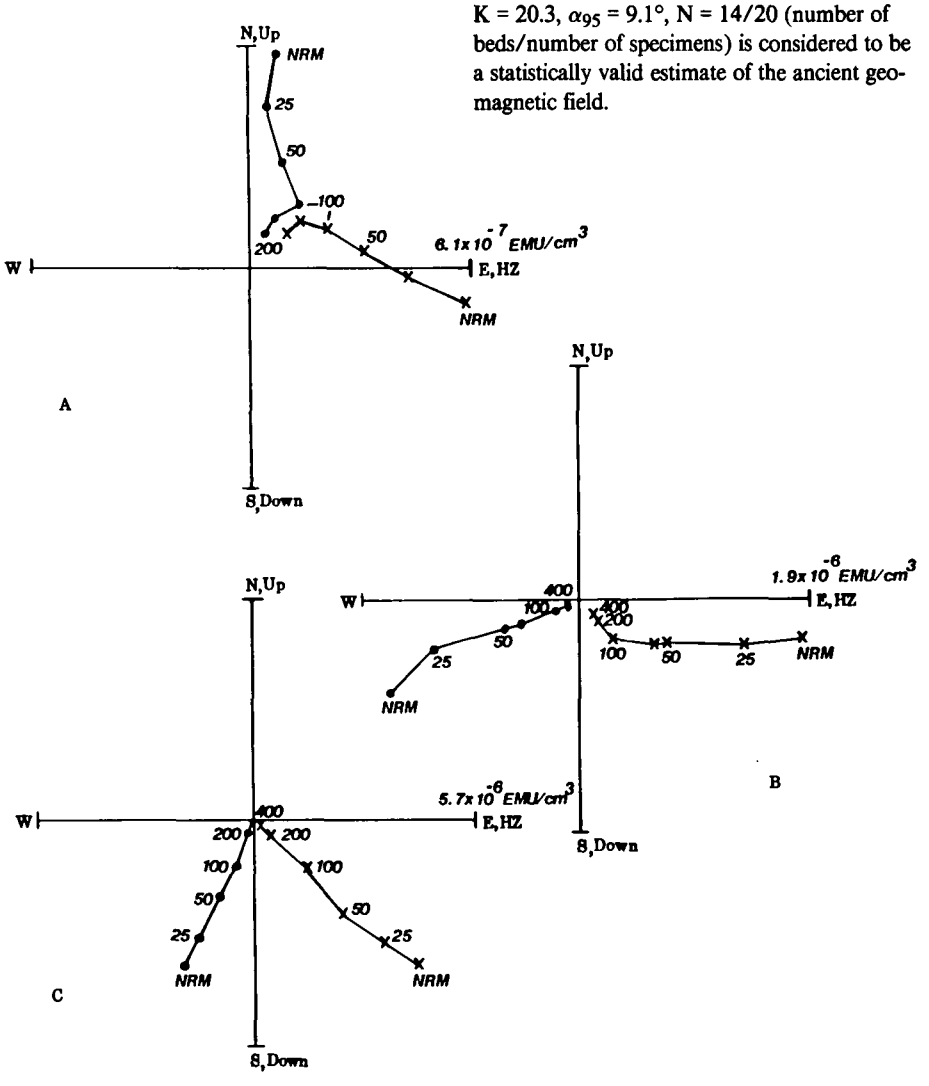


Figure 2. Demagnetization vector diagrams for three specimens from MacColl Ridge Formation. A: Bed 11, WRG.22. B: Bed 7, WRG.21. C: Bed 12, WRG.21. Dots represent declination; crosses represent inclination. Demagnetization intensity is in oersteds.

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TABLE 1. CHARACTERISTIC DIRECTIONS OF MAGNETIZATION IN MACCOLL RIDGE FORMATION

Bed	Demag Range*	N	Geographic Dec	Geographic Inc	Geographic K	Stratigraphic Dec	Stratigraphic Inc	Stratigraphic K	VGP Long	VGP Lat	VGP K
WRG.21; Lat 61.18°N, Long 217.65°											
1	100-150	2	343	76	12	241	49	11	169	14	9
4	200	3	357	84	10	216	51	10	185	12	6
5	200	1	95	-78 R	--	45	-40 R	--	357	-1 R	--
6	100-200	2	22	68	268	221	68	243	191	27	130
7	100	1	341	73	--	239	54	--	170	18	--
9	100	1	48	87	--	203	52	--	198	6	--
12	200	1	128	82	--	201	43	--	199	-2	--
	Mean		355	82		221	52		183	11	
	K =		49.5			36.3			22.8		
	α 95 =		8.7			10.2			12.9		
	N = 7/11		Approximate strike and dip of bedding = N 85° W, 45° S								
WRG.22; Lat 61.18°N, Long 217.64°											
1	200	1	75	-32 R	--	80	-31 R	--	324	-10 R	--
4	200	1	221	43	--	230	47	--	175	8	--
6	100-200	2	220	49	6	229	53	7	177	17	6
7	200-600	2	243	68	439	259	67	375	164	37	194
8	300	1	49	-32 R	--	53	-33 R	--	348	0 R	--
9	300	1	181	64	--	191	70	--	210	25	--
11	200	1	17	-32 R	--	21	-36 R	--	18	7 R	--
	Mean		222	48		231	50		177	14	
	K =		13.1			13.2			10.1		
	α 95 =		17.3			17.3			20.0		
	N = 7/9		Approximate strike and dip of bedding = N 85° W, 18° N								

Note: K is Fisher precision parameter; α_{95} is circle of 95% confidence; N is the number of beds/specimens. Missing beds were eliminated because no magnetically stable demagnetization endpoints could be isolated.

* Intensity or range of intensities (in oersteds) that isolated characteristic directions of magnetization.

DISCUSSION

The mean inclination of the MacColl Ridge Formation data corresponds to a paleolatitude of 32°N, with about 9° of analytical uncertainty at the 95% confidence level. The Northern Hemisphere position is preferred on the basis of minimum required terrane motion. If Wrangellia were in its present location with respect to North America, the predicted paleolatitude would be about 76°N. The observed paleolatitude is considerably lower than this value. Thus, the paleomagnetic evidence strongly suggests that Wrangellia and the entire Southern Alaska superterrane were located at least 3500 km south of their present locations in latest Cretaceous time and that accretion to North America did not occur until some time in the Tertiary.

There are a few paleomagnetic studies that bear on the final age of emplacement. Stone and Packer (1979) presented evidence that the Peninsular terrane was slightly south of its present location in Eocene time, although the error limits would allow the terrane to be at its present latitude. Elsewhere in the Peninsular terrane, a paleomagnetic study (Stone, 1984) suggested a low paleolatitude in Paleocene time. These data pass all the stability tests that could be applied, although the primary nature of the remanence is still equivocal.

Hillhouse and Gromme (1983) conducted a study on 50 Ma volcanic rocks from the northern edge of Wrangellia, which yielded a paleolatitude of 76°N. Their findings suggest that Wrangellia was emplaced against North America by Eocene time. If this conclusion is correct, it implies that Wrangellia (and affiliated terranes) traveled from a 32°N latitude in Maas-

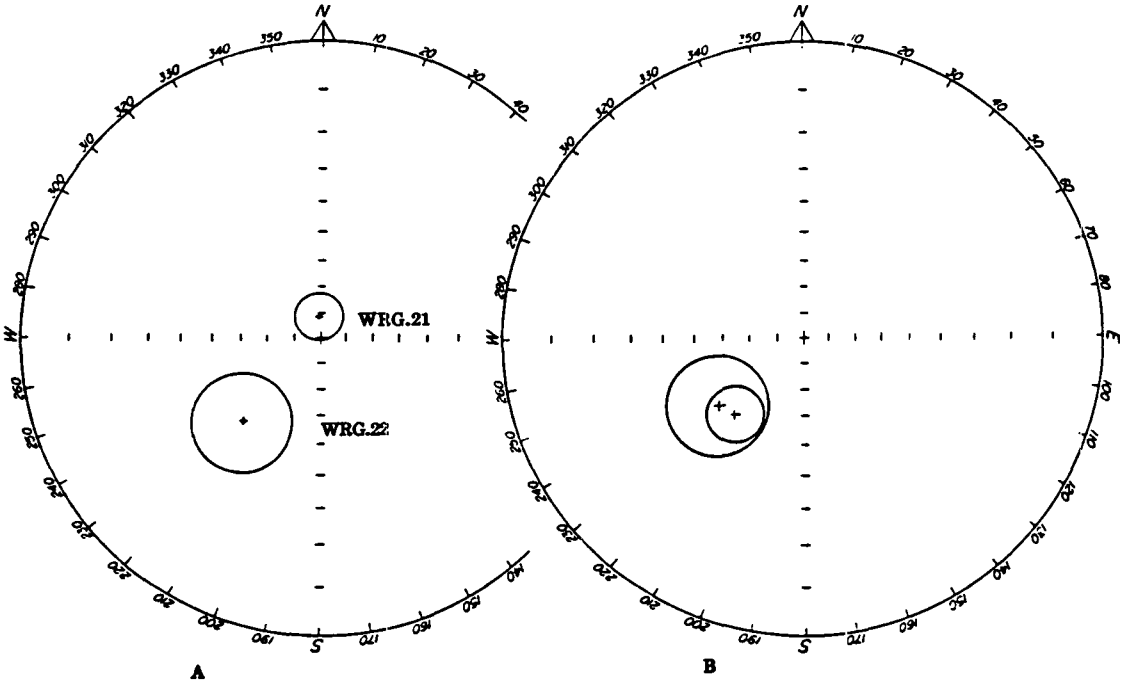


Figure 3. Mean paleomagnetic vector directions for two limbs of MacColl Ridge syncline, WRG.21 (northern limb) and WRG.22 (southern limb). A: Mean geographic vector differences (before tilt correction) showing statistical distinctness. B: Mean stratigraphic vector directions showing excellent agreement after applying tilt correction. This constitutes a positive fold test, proving that magnetization of rocks was acquired before folding.

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trichtian time, nearly 5000 km northward during the next 20 m.y. To do this would require a minimum plate motion of about 24 cm/yr (the northward component of plate velocity), which would rival the fastest known plate motion. An alternative to invoking such a large plate velocity is to postulate an unrecognized major tectonic boundary between the Hillhouse and Gromme (1983) sampling locality along the northern flank of Wrangellia and the more southerly parts of the terrane.

The mean paleomagnetic declination of the MacColl Ridge Formation is 226°. This implies that at least the MacColl Ridge area of Wrangellia rotated about 130° in a counterclockwise sense since latest Cretaceous time. This contrasts markedly with the observation of Cenozoic clockwise rotations for most of western North American displaced terranes. Alternatively, the direction of magnetization could be explained by 230° clockwise rotation; however, the near doubling of the amount of rotation seems rather extreme. The counterclockwise rotation might be accounted for by some accretionary oroclinal bending rather than rotation in a right-lateral strike-slip system. In any case, meaningful models incorporating tectonic rotations can not be constructed at present, owing to the meager data base.

The low paleolatitude determined for Wrangellia in the latest Cretaceous is rather close to the Paleocene 40° paleolatitude for the Prince William terrane reported by Plumley et al. (1983). This result lends support to the concept that the Southern Alaska superterrane, the Prince William terrane, and the intervening Chugach terrane were part of the same paleogeographic entity since early Tertiary time. Since paleolongitude is generally indeterminate by paleomagnetic data alone, the similarity in the paleolatitude determinations for Wrangellia and the Prince William terranes does not, by itself, constitute proof of paleogeographic ties. It is, however, interesting to note that many authors have provided data suggesting geologic ties between either Wrangellia and the Chugach terrane or the Chugach and the Prince William terranes by early Tertiary time or earlier (Pavlis, 1982; Moore et al., 1983; Little, 1984; Fisher and von Huene, 1984).

Given the magnitude of the tectonic implications of these data and in view of the high sample-rejection rate (based on observed directional instability during cleaning), one might question the reliability of the data. The positive fold tests and the presence of stratigraphically controlled reversals provide powerful evidence for a primary magnetic remanence. Although more data are desirable for better statistical constraint of the mean paleomagnetic direction, it is very unlikely that this mean would change enough to alter the conclusion of significant northward displacement since Cretaceous time.

CONCLUSIONS

Paleomagnetic data from the latest Cretaceous MacColl Ridge Formation yield a paleolatitude of 32° ±9°, probably in the Northern Hemisphere. This observed latitude is far south of its predicted position with respect to North America. These data strongly suggest a Cenozoic emplacement age for Wrangellia and all geologically linked terranes composing the Southern Alaska superterrane. In addition, the anomalous paleomagnetic declination can best be interpreted as the result of about 130° of Cenozoic counterclockwise rotation.

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